

Tech trends and Analysis

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OUTLINE



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EXECUTIVE SUMMARY



- Analyze tech industry trends focusing on:
 - Technology usage patterns
 - Current employment conditions
 - Worker preferences and expectations
- Generate actionable insights for:
 - Workforce development strategies
 - Technology adoption decisions
 - Organizational policy updates
- Key Goals:
 - Identify current technology usage trends
 - Understand employment conditions across the industry
 - Map worker preferences in the tech sector



INTRODUCTION



- Purpose:
 - Analyze tech industry survey data to identify trends in: • Technology usage • Employment conditions • Worker preferences
 - Provide actionable insights for strategic decisions in: • Workforce development • Technology adoption • Organizational policies
- Target Audience:
 - Tech Industry Leaders
 - HR Professionals
 - Technology Trainers/Educators
 - Policymakers
- Value Proposition:
 - Current Insights • Popular technologies and frameworks
 - Employee Satisfaction Analysis • Job satisfaction and compensation trends
 - Future Trends • Desired platforms and work arrangements\



METHODOLOGY



- Data Source:
 - Primary Dataset: survey_data_updated.csv • Tech industry responses • Technology usage data • Employment conditions • Job satisfaction metrics • Compensation information • Demographic details
- Collection Methods:
 - Online Survey Platform • Voluntary responses
- Data Processing Steps:
 - Data Loading • Pandas DataFrame implementation
 - Data Cleaning • Handled missing values • Removed incomplete responses • Converted data types
 - Analysis Preparation • Applied relevant filters • Created top 10 rankings • Sorted by frequency • Excluded null values



RESULTS

Technology Trends

- Top Programming Languages • Leaders: Python, JavaScript, Java
- Popular Web Frameworks • Primary: React, Angular

Employment Insights

- Job Satisfaction • Highest in full-time roles •
- Compensation Patterns • Peak: Ages 31-40 • Median: \$85,000/year

Demographics & Education

- Age Distribution • Core: 25-34 age group • Diverse workforce range
- Educational Background • Majority: Bachelor's or higher

Future Trends & Preferences

- Emerging Platforms • Focus: Cloud computing, AI
- Work Arrangements • 60% prefer remote work

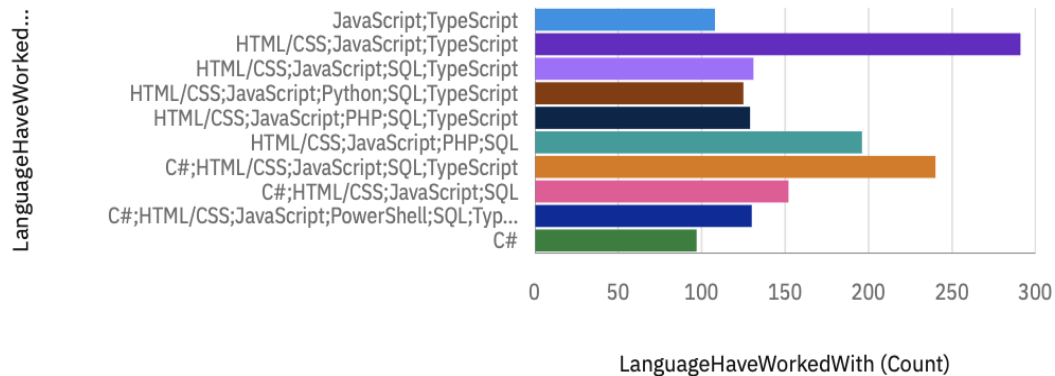
PROGRAMMING LANGUAGE TRENDS

Current Year

Top 10 LanguageHaveWorkedWith

LanguageHaveWorkedWith

- C#
- C#;HTML/CSS;JavaScript;SQL
- HTML/CSS;JavaScript;PHP;SQL
- HTML/CSS;JavaScript;Python;SQL
- C#;HTML/CSS;JavaScript;PowerS...
- C#;HTML/CSS;JavaScript;SQL;Ty...
- HTML/CSS;JavaScript;PHP;SQL;T...
- HTML/CSS;JavaScript;SQL;TypeS...

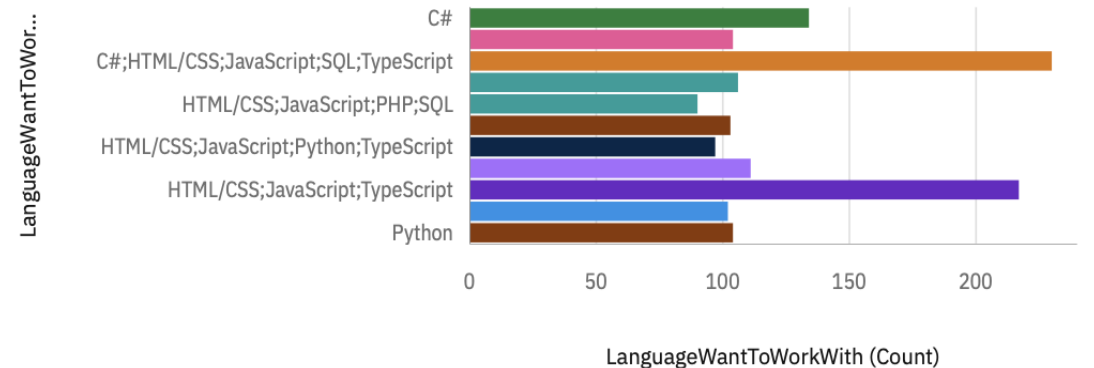


Next Year

Top 10 LanguageWantToWorkWith

LanguageWantToWorkWith

- C#
- C#;HTML/CSS;JavaScript;SQL
- C#;HTML/CSS;JavaScript;SQL;Ty...
- HTML/CSS;JavaScript;PHP;SQL
- HTML/CSS;JavaScript;Python;SQL
- HTML/CSS;JavaScript;Python;Type...
- HTML/CSS;JavaScript;SQL;TypeS...



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Python, JavaScript, and Java emerged as the most popular languages among respondents.
- Respondents are keen to work with emerging platforms such as cloud computing and AI
- Respondants wants to use C# more in their future tech stack

Implications

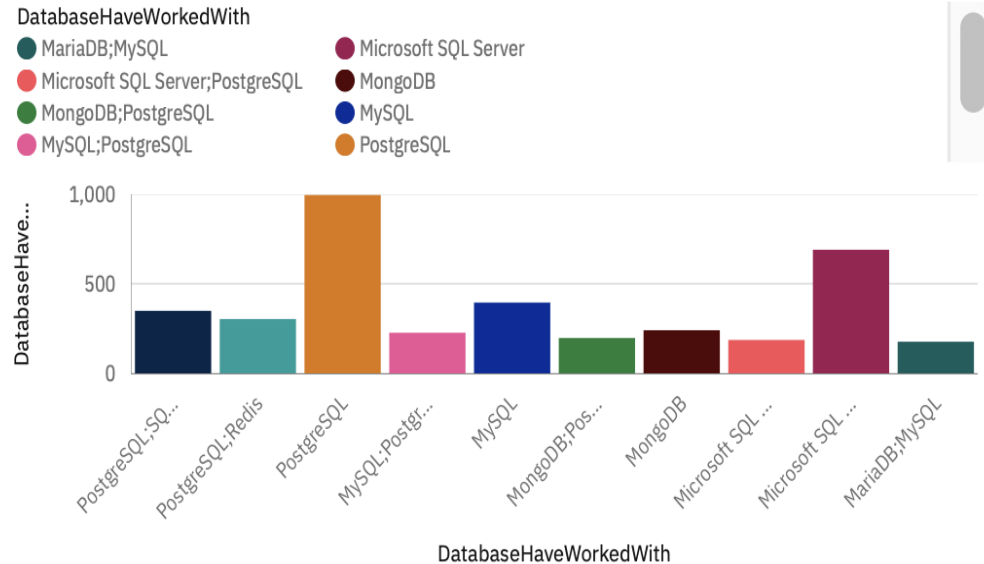
- These languages dominate the tech landscape, reflecting their widespread use in various applications
- These platforms represent the future of technology innovation and development
- Adoption of New Languages can help organizations stay ahead of the curve by adopting new technologies early.



DATABASE TRENDS

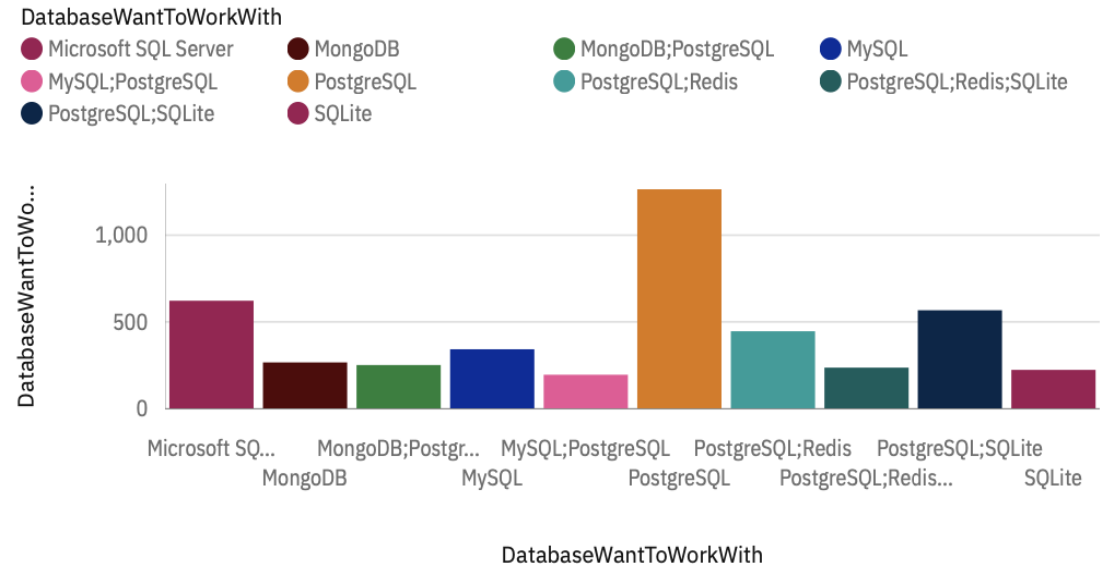
Current Year

Top 10 DatabaseHaveWorkedWith



Next Year

Top 10 DatabaseWantToWorkWith



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most desired database to work with, significantly leading with over 1,000 mentions.
- Microsoft SQL Server is the second most popular choice, with around 600 mentions.
- There is a notable interest in combinations of databases, such as MongoDB with PostgreSQL and PostgreSQL with Redis, indicating a trend towards using multiple databases in tandem.

Implications

- The high interest in PostgreSQL suggests that there will be a growing demand for professionals skilled in PostgreSQL.
- The interest in Microsoft SQL Server implies that it is a database with strong demand for future use.
- Companies providing database solutions might need to focus on interoperability and integration features to cater to the trend of using multiple databases together.



DASHBOARD



https://github.com/wisdomidi/dashboard/blob/main/survey_data%20dashboard.pdf

DASHBOARD TAB 1

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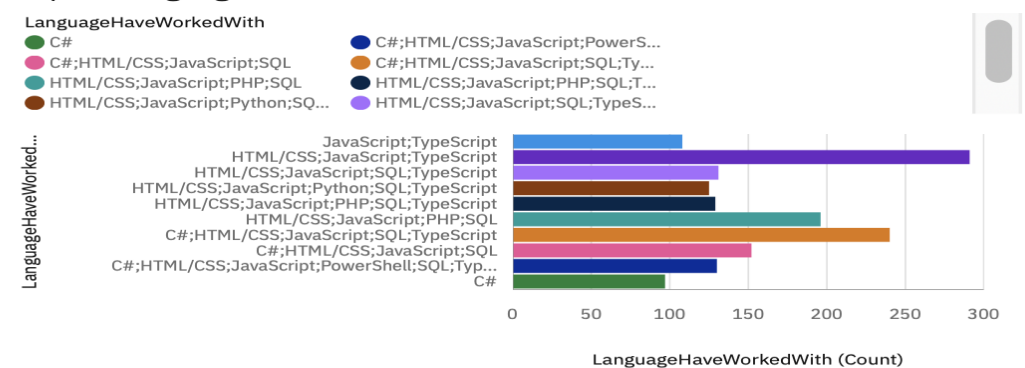
survey_data dashboard

Current Technology Usage

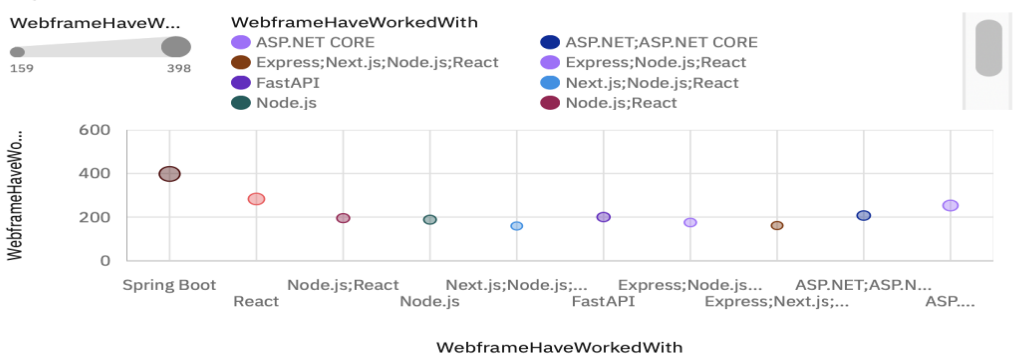
Top 10 PlatformHaveWorkedWith



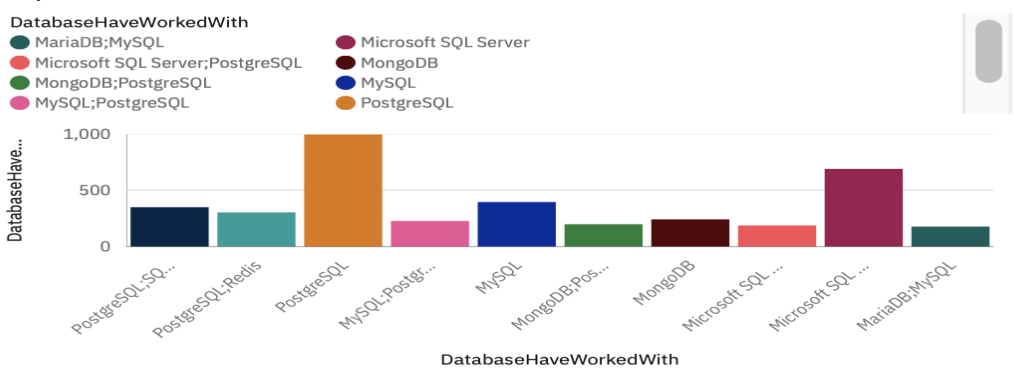
Top 10 LanguageHaveWorkedWith



Top 10 WebFrameHaveWorkedWith



Top 10 DatabaseHaveWorkedWith

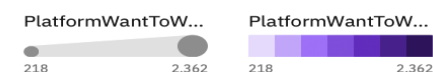


DASHBOARD TAB 2

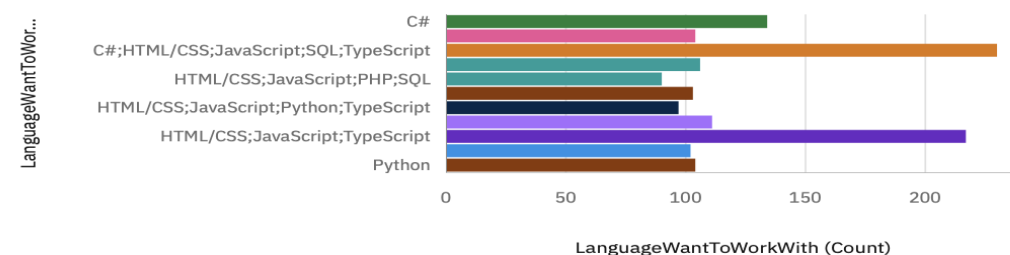
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Future Technology Trend

Top 10 PlatformWantToWorkWith

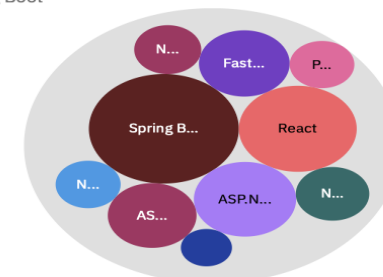
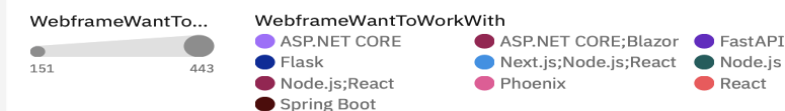


Top 10 LanguageWantToWorkWith

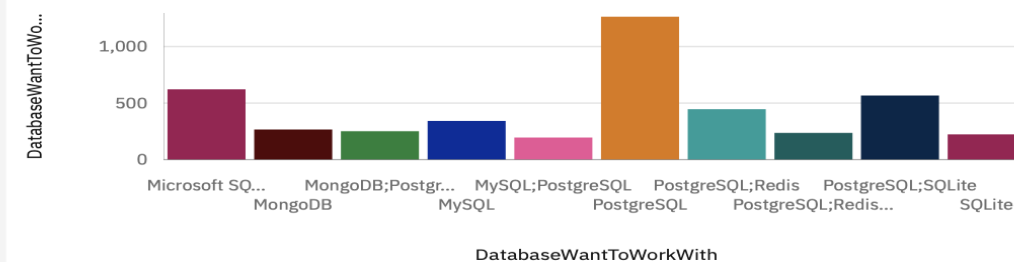


survey_data dashboard

Top 10 WebframeWantToWorkWith



Top 10 DatabaseWantToWorkWith

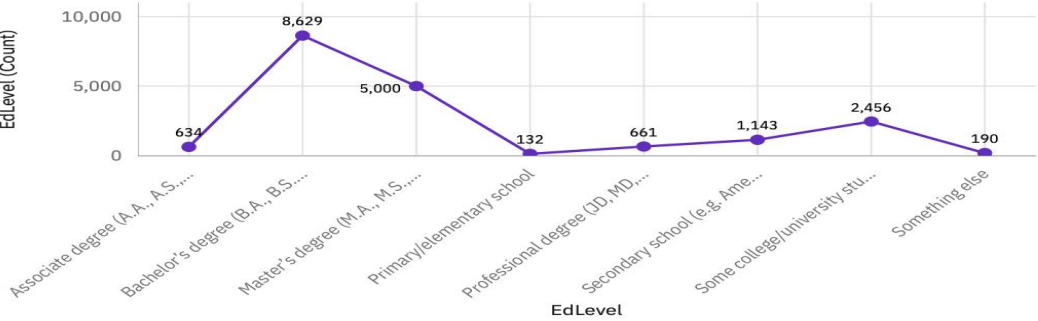


DASHBOARD TAB 3

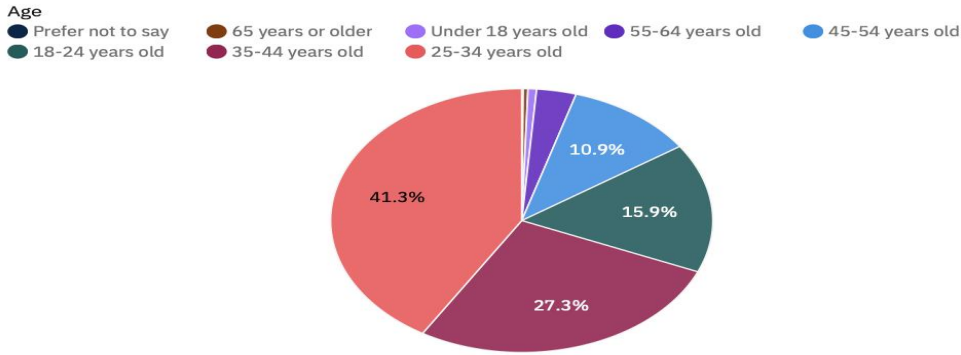
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Demographics

Respondent distribution by Formal Education Level

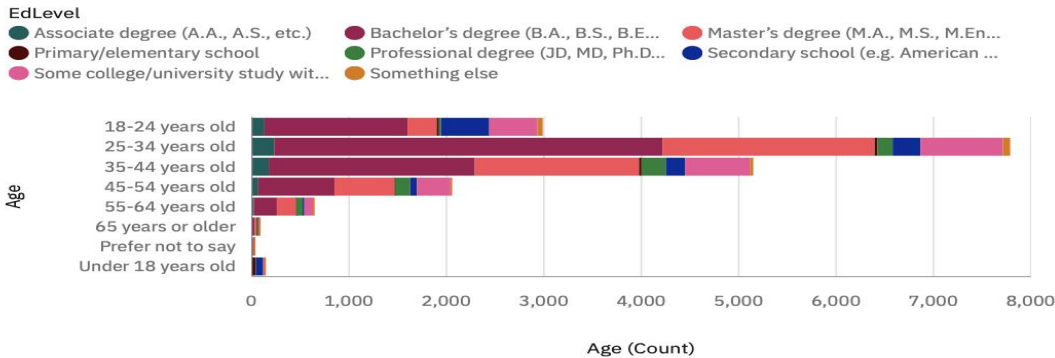


Respondent distribution by Age

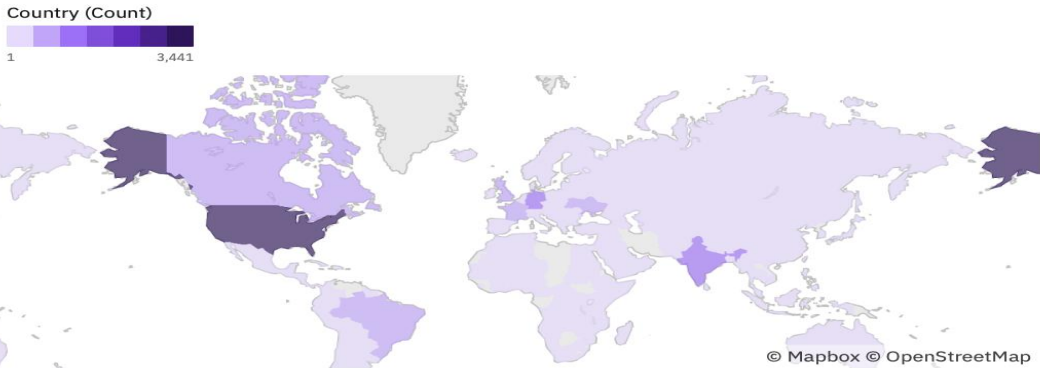


survey_data dashboard

Respondent Count by Age, classified by Education Level



Respondent Count by Country



DISCUSSION



- The analysis revealed key trends in technology preferences, employment conditions, and demographics within the tech industry.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Python and JavaScript are the most popular programming languages.
- 60% of respondents prefer remote work, indicating a trend towards flexible work arrangements.
- Full-time employees report higher job satisfaction compared to part-time and freelance workers.

Implications

- Invest in training programs for Python and JavaScript
- Promote remote work to attract and retain talent, aligning with the strong preference for flexible work arrangements.
- Encourage continuous learning to improve job satisfaction and retention.



CONCLUSION



- Enhanced workforce development through targeted training programs.
- Informed technology adoption by aligning with industry trends.
- Improved employee satisfaction and retention by addressing job satisfaction factors.
- Strategic recruitment aligned with in-demand skills and languages.
- Promotion of flexible work arrangements to meet employee preferences.



APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

